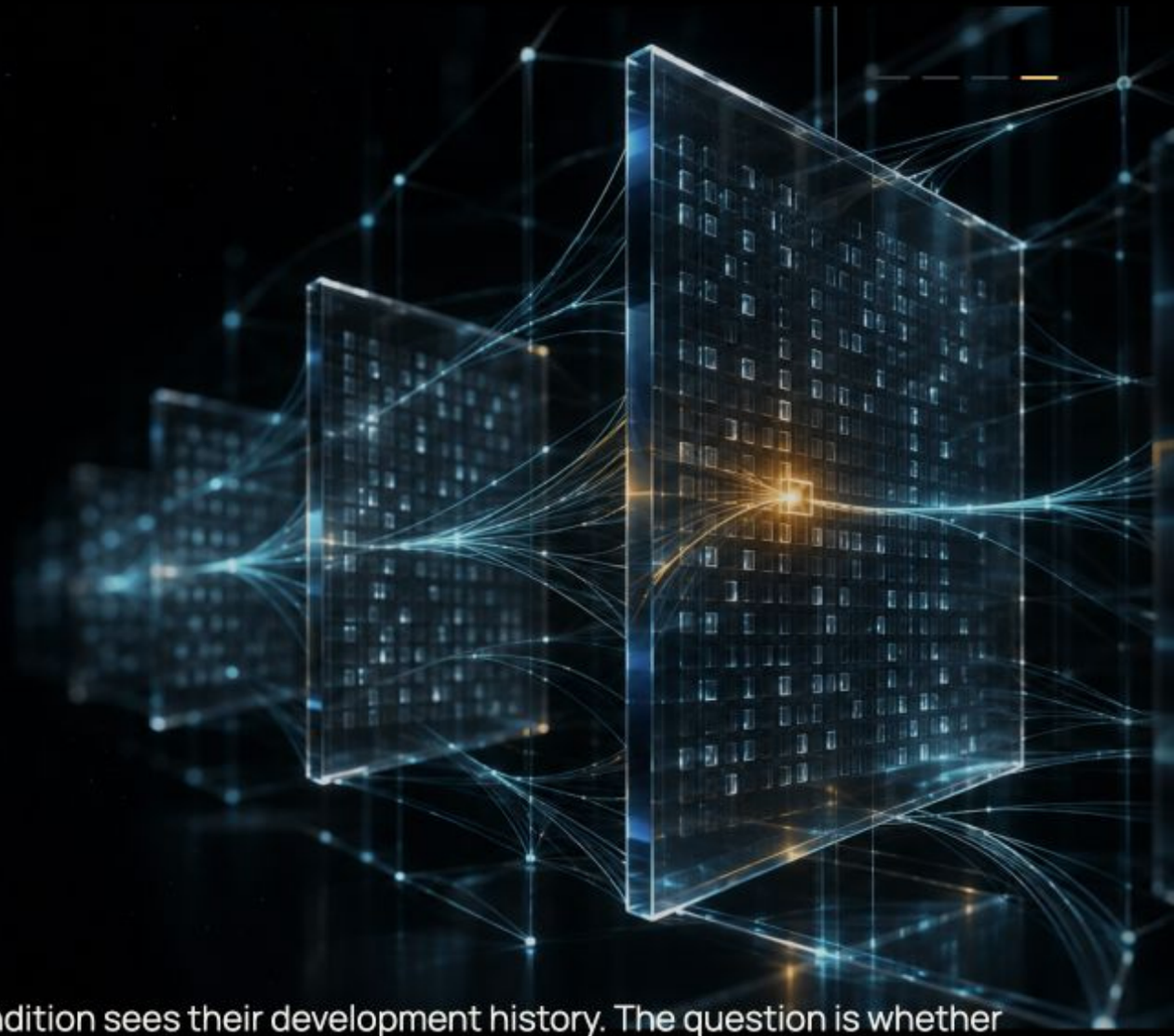
Editing Neural Networks with Language Models

Independent or together?



TWO TARGETS

Census-income classifier
Intact language model

Independent / cooperative

Two configured editors search independently, then a joint condition sees their development history. The question is whether those changes improve performance on frozen test panels.

Begin with the actual article

Sparks of Weight-Space Programming



Sparks of Weight-Space Programming:

LLMs can natively read & write neural network weights

AUTHORS	AFFILIATIONS	PUBLISHED
Rikiya Takehi*	MIT CSAIL	November 2, 2026
Jayden Teoh*	MIT CSAIL	
Alex Zhang	MIT CSAIL	
Omar Khattab	MIT CSAIL	

Rikiya Takehi · Jayden Teoh · Alex Zhang · Omar Khattab
2 November 2026 · Research blog article

Rikiya Takehi, Jayden Teoh, Alex Zhang and Omar Khattab propose that language models can natively read and write neural network weights. This is their research blog article.

Begin with the actual article

Reading and writing parameters



Sparks of Weight-Space Programming:

LLMs can natively read & write neural network weights

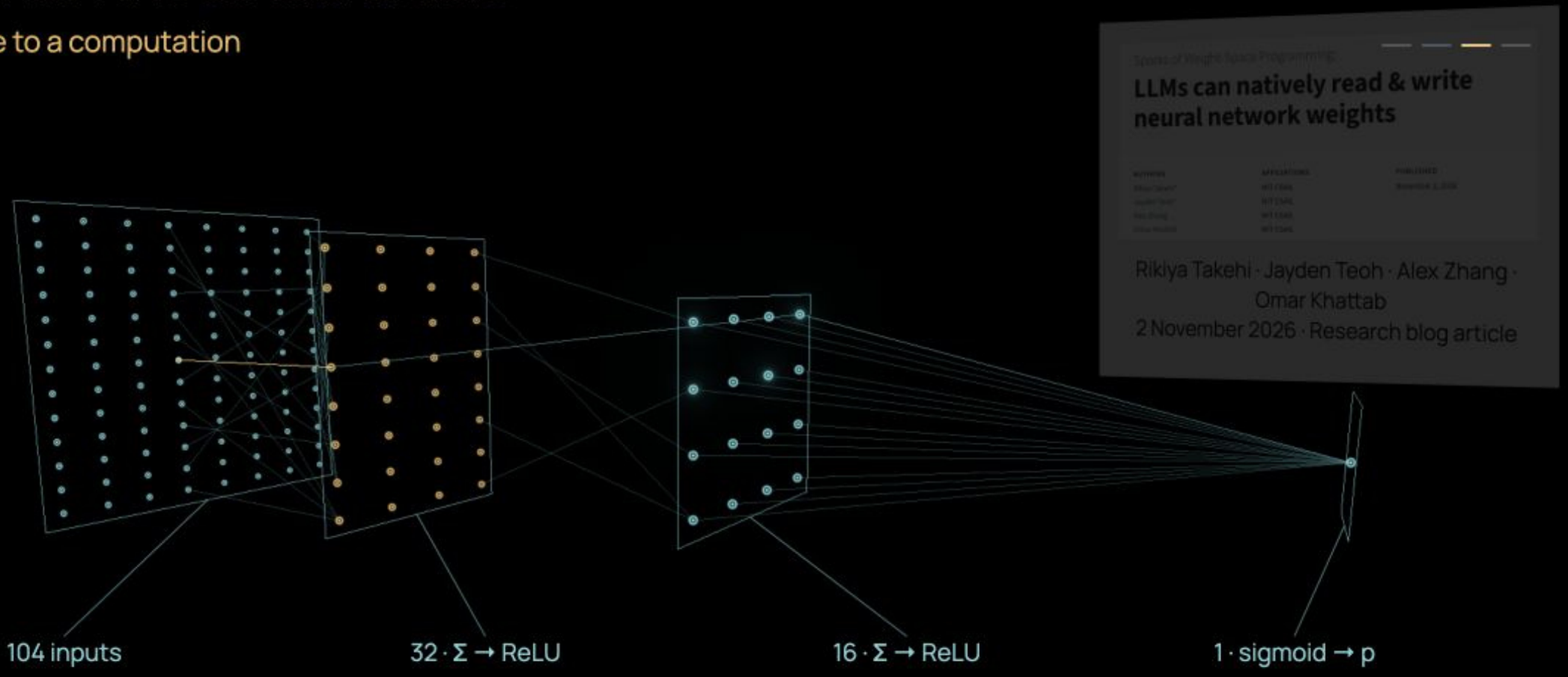
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Omar Khattab	MIT CSAIL	

Rikiya Takehi · Jayden Teoh · Alex Zhang · Omar Khattab
2 November 2026 · Research blog article

Weight-space programming treats numerical parameters as objects a language model can reason about and alter. The source demonstrates small policy repair as well as edits to a Qwen language model.

Begin with the actual article

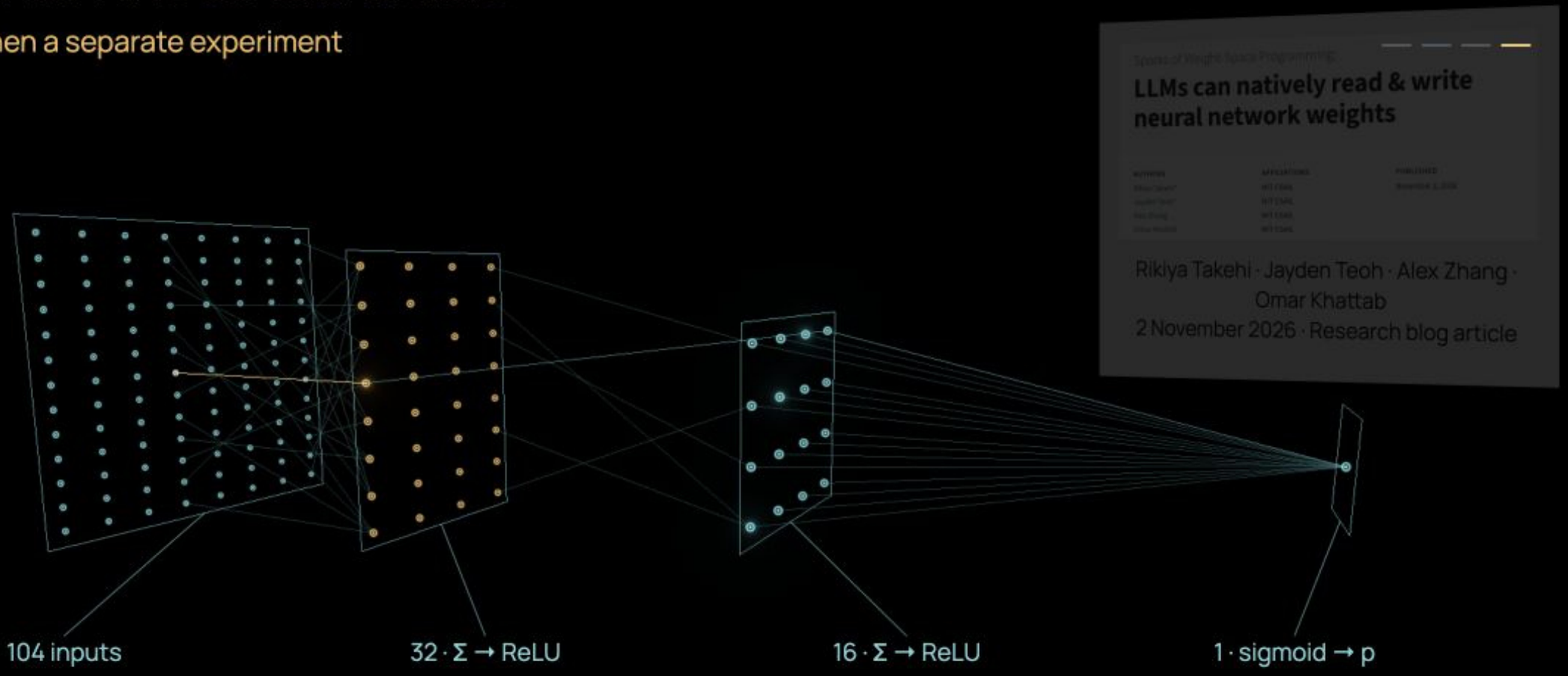
From a page to a computation



Move from the article to a network: the parameters determine how inputs become outputs. Changing those parameters can change behavior without retraining the entire model.

Begin with the actual article

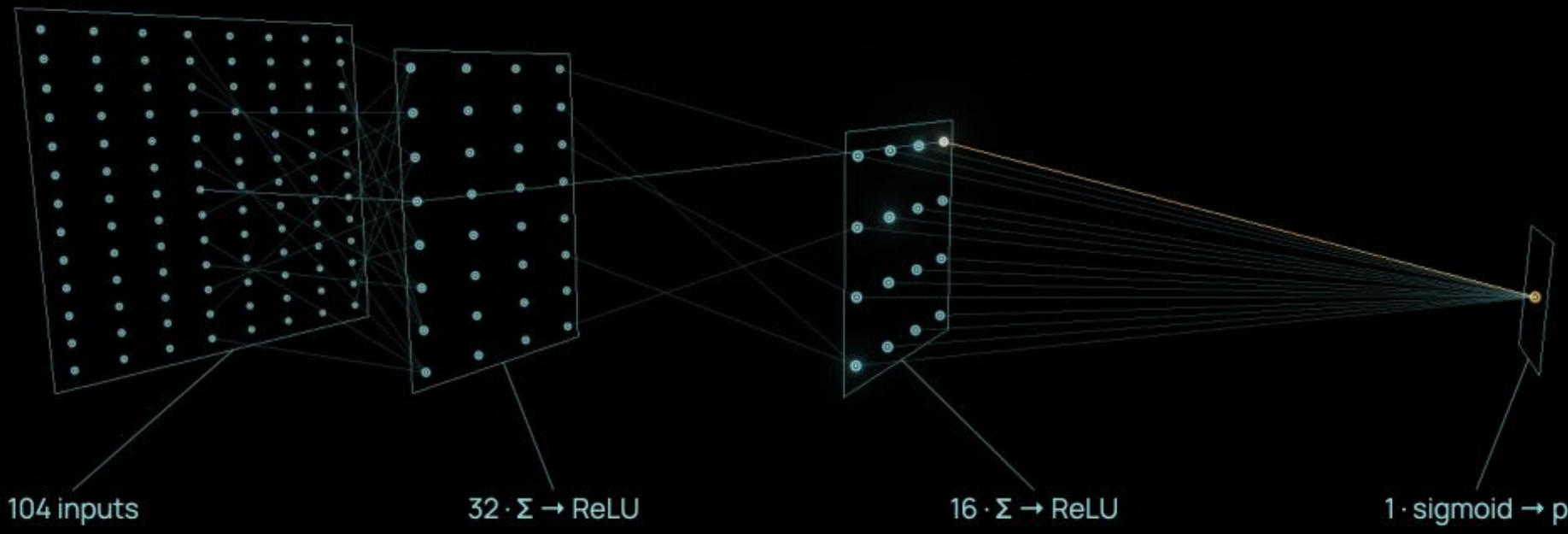
A source, then a separate experiment



The source motivates our work. Its demonstrations and outcomes remain reference-study results; the measurements that follow belong to our own bounded comparison.

What does a weight edit mean?

Evaluate without gradient descent



All 104 → 32 → 16 → 1 nodes · sampled context edges · schematic signal, not measured activations

During this editing stage there is no gradient descent. The target performs forward computations, the controller scores development predictions, and selection chooses among legal candidates.

Reference demonstrations; our question

Development-informed cooperation

Reference study results	Our experiments
Read / write weights	Damaged Adult classifier
Small policy repair	Intact SmolLM2-1.7B
Qwen example	Independent / joint selection

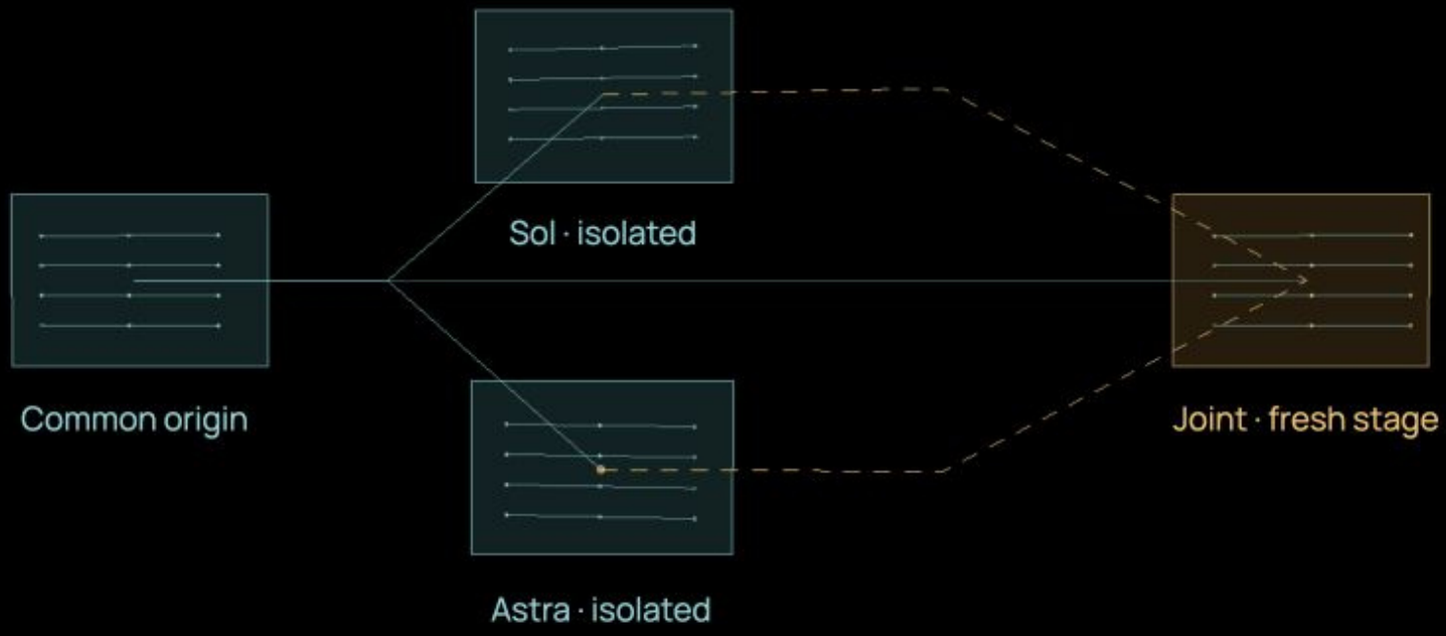
Reference demonstrations motivate our separate bounded comparison

The joint condition receives independent development histories. It still starts from the common origin and cannot see frozen test feedback when selecting its patch.

Reference demonstrations are not our results. Different targets and bounded protocols; no claim of full replication.

The experimental map

Frozen tests compare



Solid cyan: weights from origin · dashed amber: development history · joint starts fresh

Once selection is locked, the held-out panel compares the selected states. We will keep actual changes, scoring conventions and the limits of this single session visible.

The UCI Adult census-income task

Predict income above $50K

Census attributes → P(income > $50K)



Sex and race excluded from encoded predictors · no personal feature rows

Adult is the name of a census-income dataset from the UCI Machine Learning Repository. The classification task maps structured census attributes to a probability of income above fifty thousand dollars.

Our Adult experiment · original test panel n = 16,281 · 3,846 positive, 12,435 negative · no personal feature records shown.

The UCI Adult census-income task

104 encoded predictors

Census attributes → P(income > $50K)



Sex and race excluded from encoded predictors · no personal feature rows

Categorical attributes are encoded and combined with numerical inputs. The resulting vector has 104 dimensions; sex and race are excluded from the encoded predictors in this experiment.

The UCI Adult census-income task

3,846 positive examples

Original test panel · 16,281 examples



Sex and race excluded from encoded predictors · no personal feature rows

Positive means income above the threshold. There are 3,846 positive examples in the original test panel, which will matter when we interpret class-specific recall.

The UCI Adult census-income task

12,435 negative examples

Original test panel · 16,281 examples



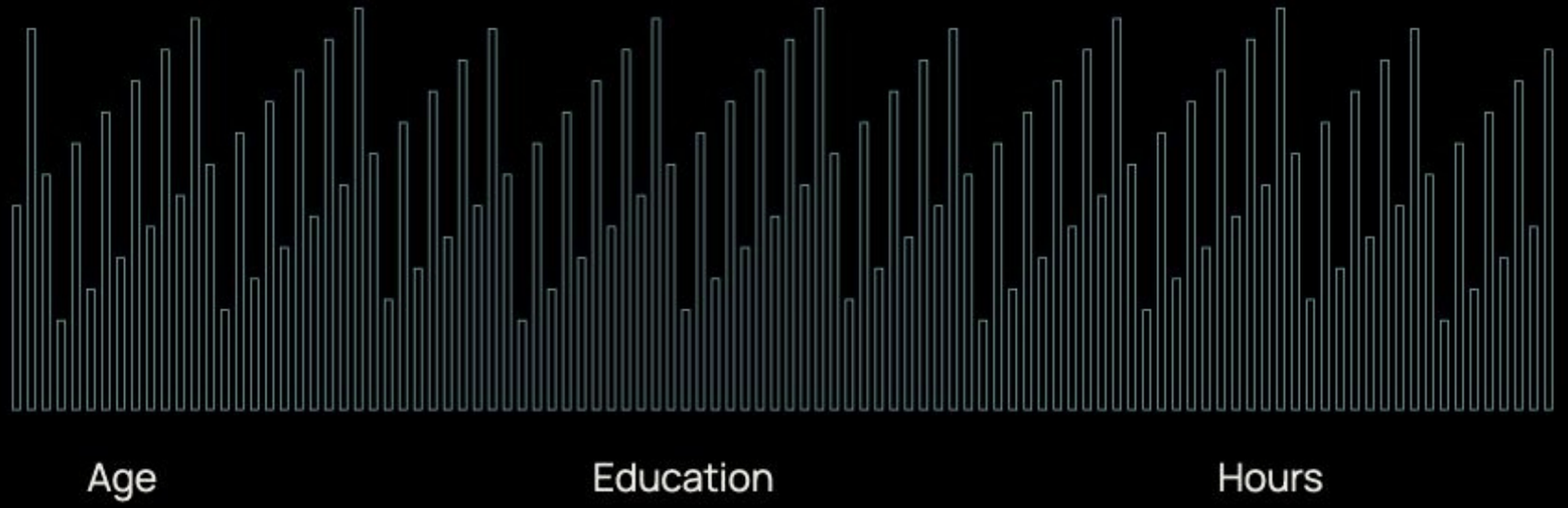
Sex and race excluded from encoded predictors · no personal feature rows

Negative means income at or below the threshold. With many more negative examples, ordinary accuracy gives their decisions more total weight than positive decisions.

Our Adult experiment · original test panel n = 16,281 · 3,846 positive, 12,435 negative · no personal feature records shown.

From features to a vector

An input is not a weight



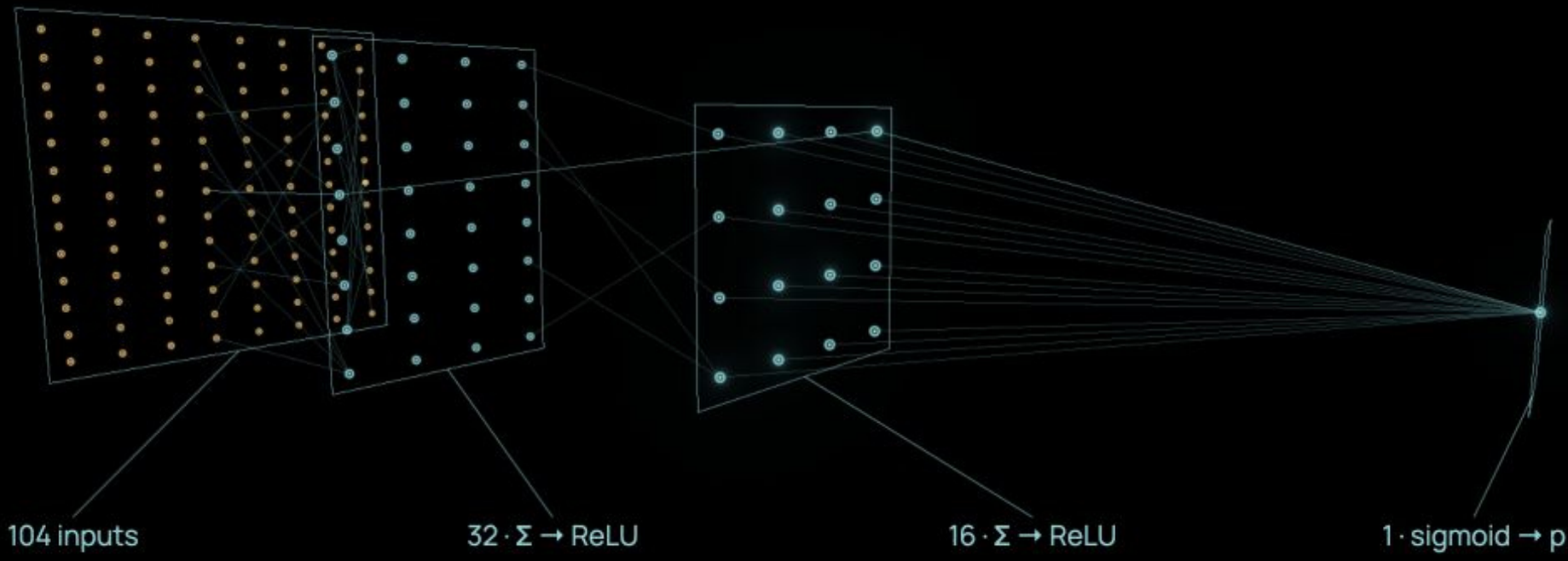
104 dimensions · schematic heights, not actual person values

The input changes from example to example. Weights are the learned multipliers applied to those coordinates, and the editing intervention changes permitted weights rather than test features.

Schematic encoding only · age / education / hours name feature categories, not a real person · 104 encoded inputs.

Follow the forward pass

104 inputs

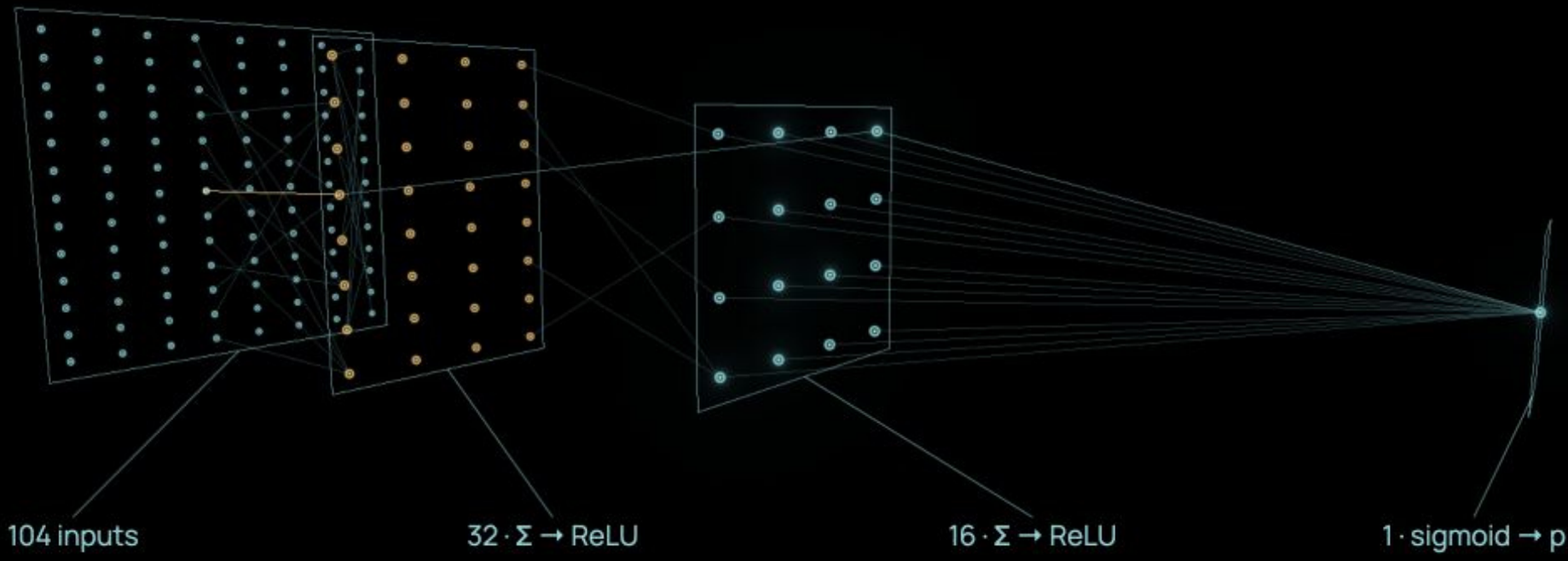


All 104 → 32 → 16 → 1 nodes · sampled context edges · schematic signal, not measured activations

The forward pass starts with 104 encoded values. Each first-layer neuron combines the input vector with its own weights and bias.

Follow the forward pass

32 rectified activations

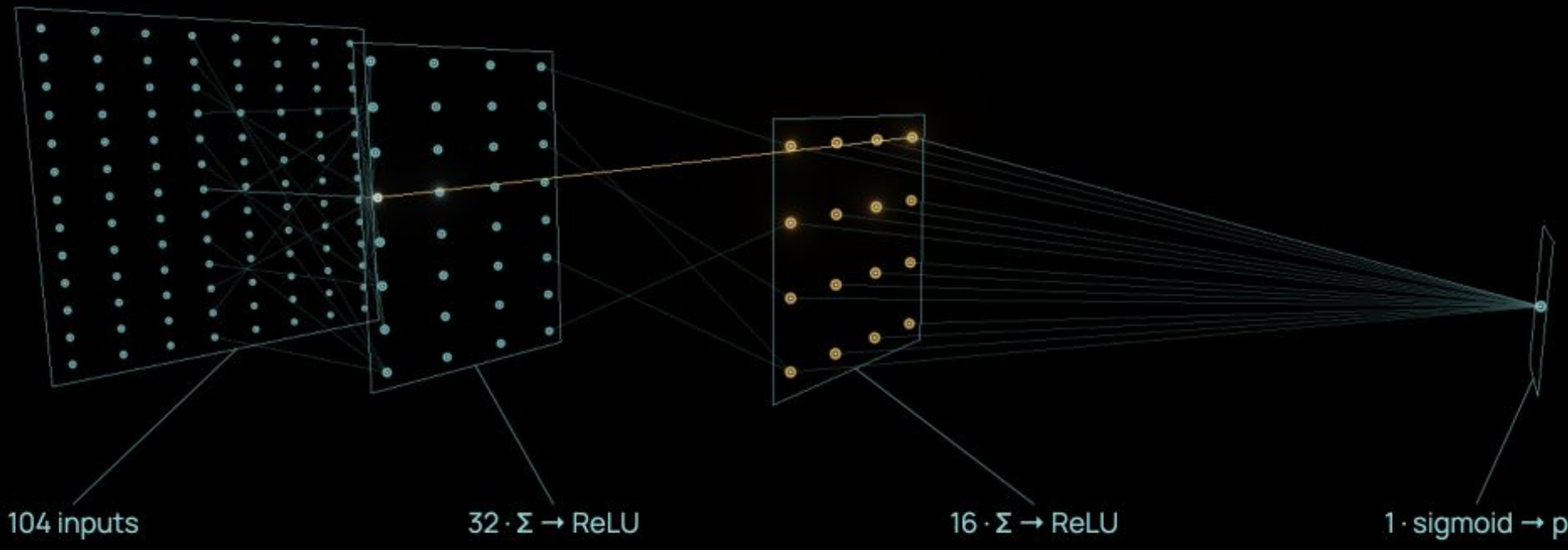


All 104 → 32 → 16 → 1 nodes · sampled context edges · schematic signal, not measured activations

The first hidden layer produces 32 values after a rectified linear unit. This activation keeps a positive weighted sum and replaces a negative weighted sum with zero.

Follow the forward pass

16 rectified activations

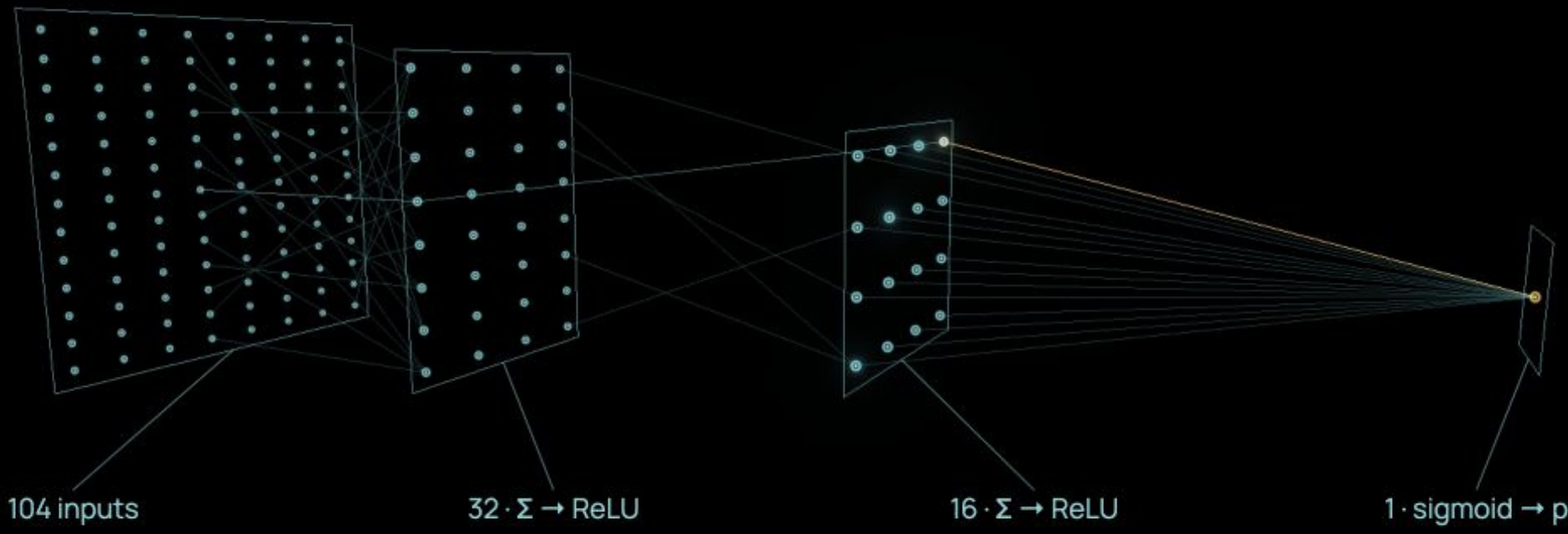


All 104 → 32 → 16 → 1 nodes · sampled context edges · schematic signal, not measured activations

The second hidden layer combines those 32 values into 16 activations. Its 16-by-32 weight matrix is the editable surface after a permanent lesion is applied.

Follow the forward pass

One sigmoid probability



All 104 → 32 → 16 → 1 nodes · sampled context edges · schematic signal, not measured activations

The final layer maps the 16 activations to one logit, an unrestricted real score. A sigmoid converts that score into a probability; classification uses a threshold of one half.

Inside one neuron

Input x · weight w · bias b



x · input, w · weight, b · bias

First-layer neuron 1

Weight × input
-0.0789 × -0.3333 = 0.02629

Weight × input
0.0598 × 0.2333 = 0.01395

Weight × input
0.0272 × -0.2333 = -0.00634

Other 101 products; bias
0.14573; 0.04632

Weighted sum z; activation h
0.22594; 0.22594

Real frozen weights · synthetic input illustration

For one neuron, x is an incoming value, w is its multiplier and b is a bias added once. Index j identifies an input coordinate; the sum runs over all coordinates.

Inside one neuron

$z = \Sigma_j w_j x_j + b$



$z = \Sigma_j w_j x_j + b$

First-layer neuron 1

Weight × input
-0.0789 × -0.3333 = 0.02629

Weight × input
0.0598 × 0.2333 = 0.01395

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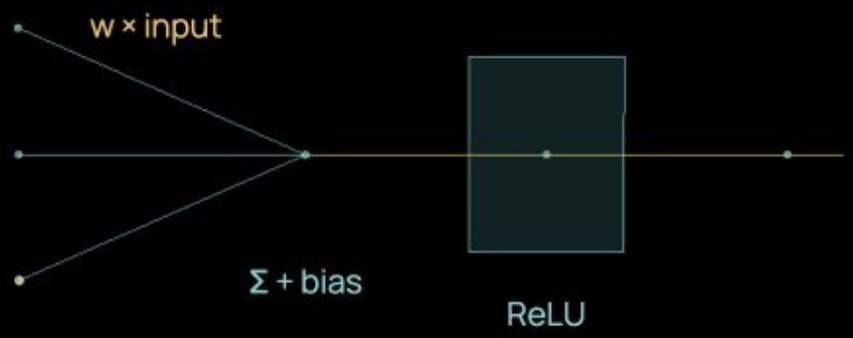
Weighted sum z; activation h
0.22594; 0.22594

Real frozen weights · synthetic input illustration

The weighted sum z adds every weight–input product and the bias. We show real frozen weights with a deterministic synthetic input so the arithmetic is inspectable without personal feature rows.

Inside one neuron

h = max(0, z)



h = max(0, z)

First-layer neuron 1

Weight × input
-0.0789 × -0.3333 = 0.02629

Weight × input
0.0598 × 0.2333 = 0.01395

Weight × input
0.0272 × -0.2333 = -0.00634

Other 101 products; bias
0.14573; 0.04632

Weighted sum z; activation h
0.22594; 0.22594

Real frozen weights · synthetic input illustration

The activation h is the rectified weighted sum. Positive values pass through; negative values become zero. The highlighted first-layer neuron has a positive sum in this synthetic illustration.

Inside one neuron

$p = 1 / (1 + \exp(-s))$



$p = 1 / (1 + \exp(-s))$

Final logit s
0.67891

Probability p
0.66350

Real frozen weights · synthetic input illustration

At the output, s denotes the final logit and p the predicted probability. The exponential function maps that logit smoothly into the interval between zero and one.

Inside one neuron

Follow values, not an accuracy animation

Input → sums → activations → p

Final logit s
0.67891

Probability p
0.66350

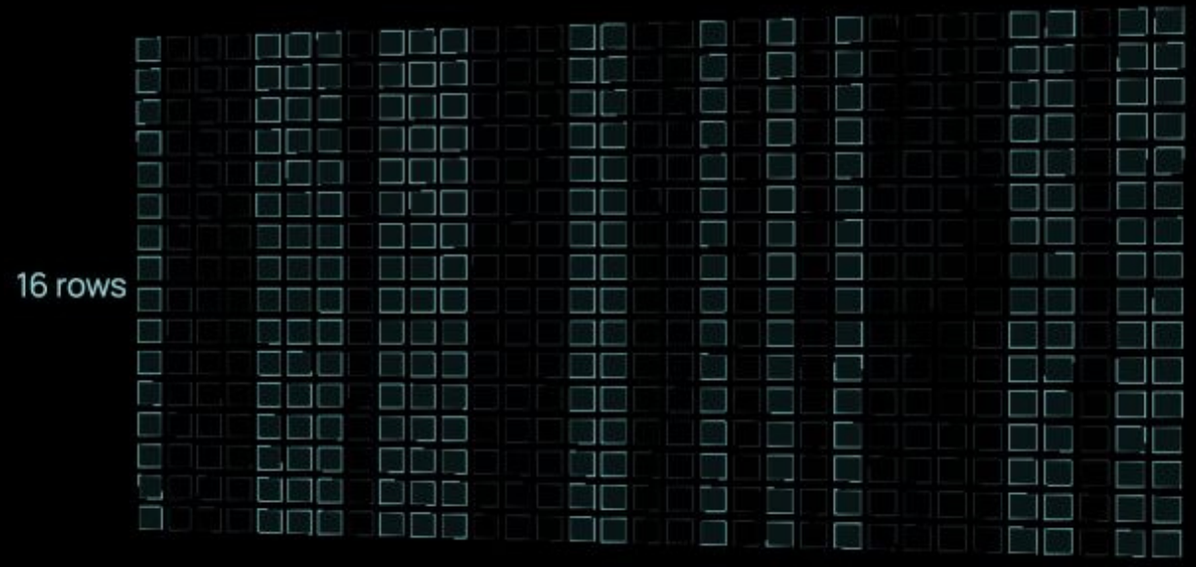


Real frozen weights · synthetic input illustration

Trace the fixed synthetic input through real retained weights. Each multiplication contributes to a sum, each rectifier gates a signal, and the final sigmoid produces the illustrated probability.

A permanent lesion

The mask survives every proposal



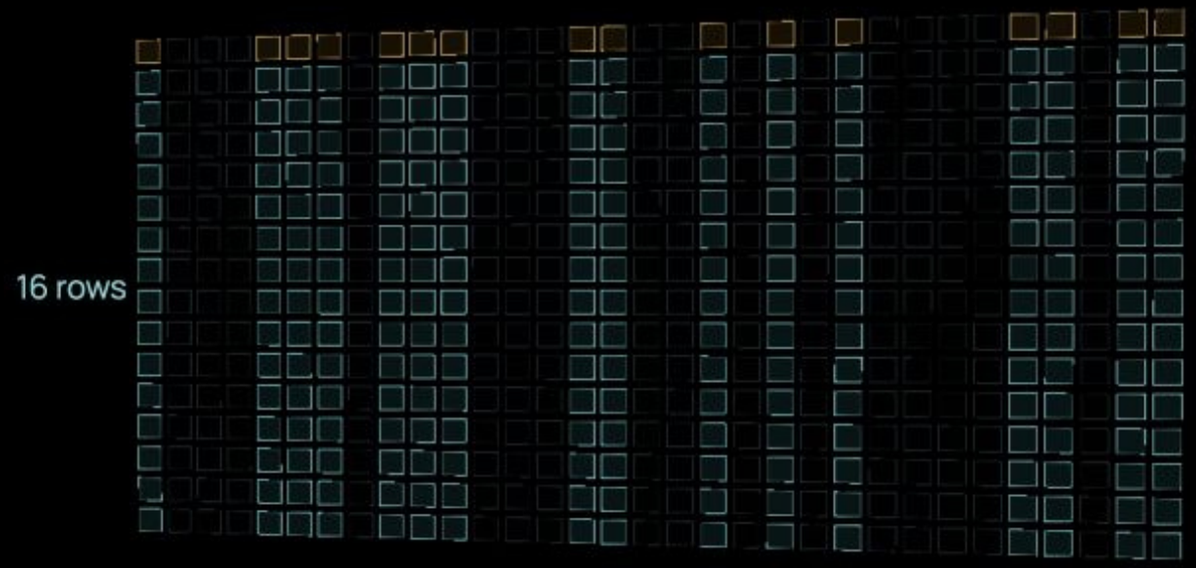
Full 16 × 32 matrix · 256 masked / 256 eligible

The controller enforces the same permanent mask after each proposal. This constraint is part of the scientific question: can changing surviving weights improve the chosen metric?

Exact frozen permanent mask · fc2.weight has 16 × 32 = 512 entries · 256 masked, 256 eligible.

What makes a patch legal?

A small, counted intervention



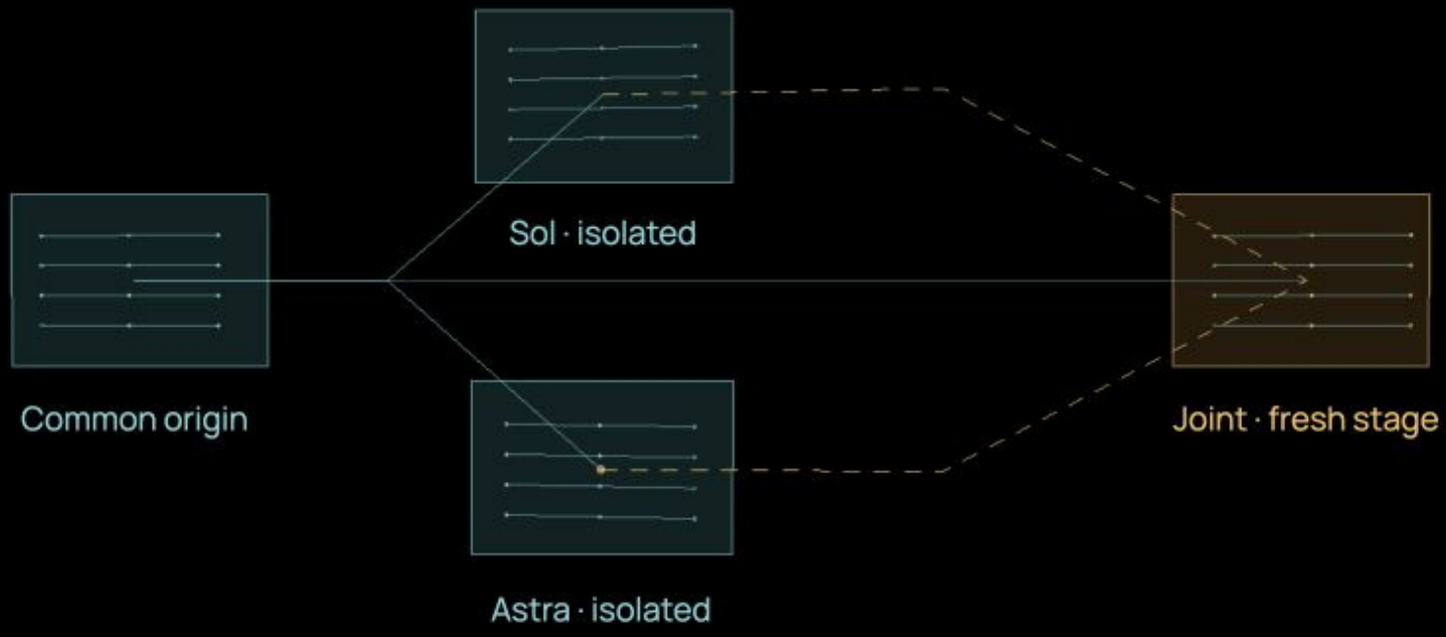
16 selected changes · relative norm 4.8918% · amber changed / cyan surviving / dark masked

The controller counts unequal executed scalars after casting, rather than counting requested operations. Both the parameter-count cap and the five-percent relative tensor norm must hold.

Adult: surviving fc2 weights only · ≤10% of model scalars · tensor relative Frobenius change ≤5% · norms after execution casting.

Who are Sol and Astra?

Joint: development history only



Solid cyan: weights from origin · dashed amber: development history · joint starts fresh

The joint stage can use the independent histories to propose fresh complete patches. This additional development context never includes final test answers.

Configured distinct LLM editor routes · medium reasoning · names identify conditions, not inferred capabilities.

Search, then freeze selection

Unequal allowances remain visible



Solo allowance 10 · joint allowance 5

The original Adult independent conditions had ten candidate attempts each; joint had five shared rounds. We retain that limitation rather than implying an equal-budget confirmation study.

Adult selection: maximize development balanced accuracy; then minimize norm, changed count, candidate index. Solo allowance 10; joint 5.

Follow one real Adult edit

Sixteen changes in row zero



Selected Sol · 16 row-0 changes · focus next: column 15

The selected Sol patch changes 16 surviving scalars in the first row of the second-layer matrix. The amber entries are actual before/after differences from the frozen graph.

Selected Sol patch · 16 actual scalar changes in row 0 · column 15 has nonzero input activation in the fixed synthetic illustration.

Follow one real Adult edit

Column 15: w × h



w × h

Weight before → after

0.12482 → 0.15282

Incoming h · column 15

0.54778

Contribution before → after

0.06837 → 0.08371

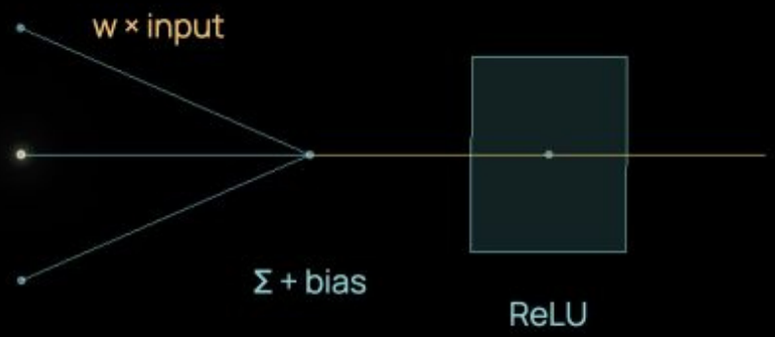
Real frozen weights · synthetic input illustration

Focus on column 15, whose incoming activation h is nonzero in our synthetic input. Its contribution to the next weighted sum is the weight w multiplied by that activation.

Selected Sol patch · 16 actual scalar changes in row 0 · column 15 has nonzero input activation in the fixed synthetic illustration.

Follow one real Adult edit

$\Delta z = \Sigma_j \Delta w_j h_j$



$\Delta z_0 = \Sigma_j \Delta w_j h_j$

Weight before → after

0.12482 → 0.15282

Incoming h · column 15

0.54778

Contribution before → after

0.06837 → 0.08371

All 16 changes · Δz_0

0.05435

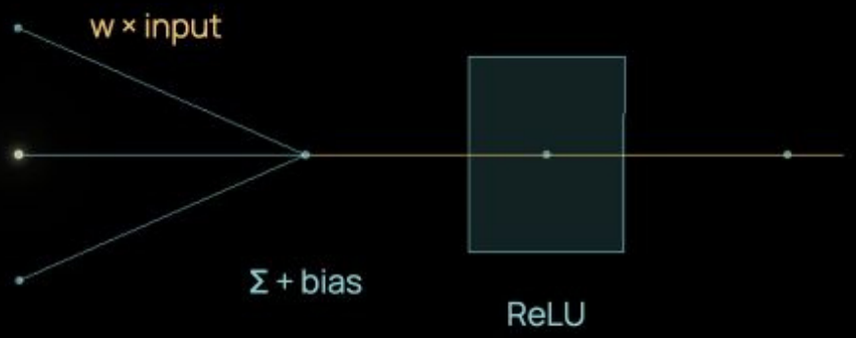
Real frozen weights · synthetic input illustration

Holding first-layer activations fixed, the change in the next weighted sum is the sum of changed-weight contributions. The highlighted product is one of those terms, not the entire effect.

Selected Sol patch · 16 actual scalar changes in row 0 · column 15 has nonzero input activation in the fixed synthetic illustration.

Follow one real Adult edit

From a changed sum to a probability



Edited sum → ReLU → sigmoid

Weight before → after
0.12482 → 0.15282

Incoming h · column 15
0.54778

Contribution before → after
0.06837 → 0.08371

All 16 changes · Δz_0
0.05435

Propagated probability
0.66350 → 0.66671

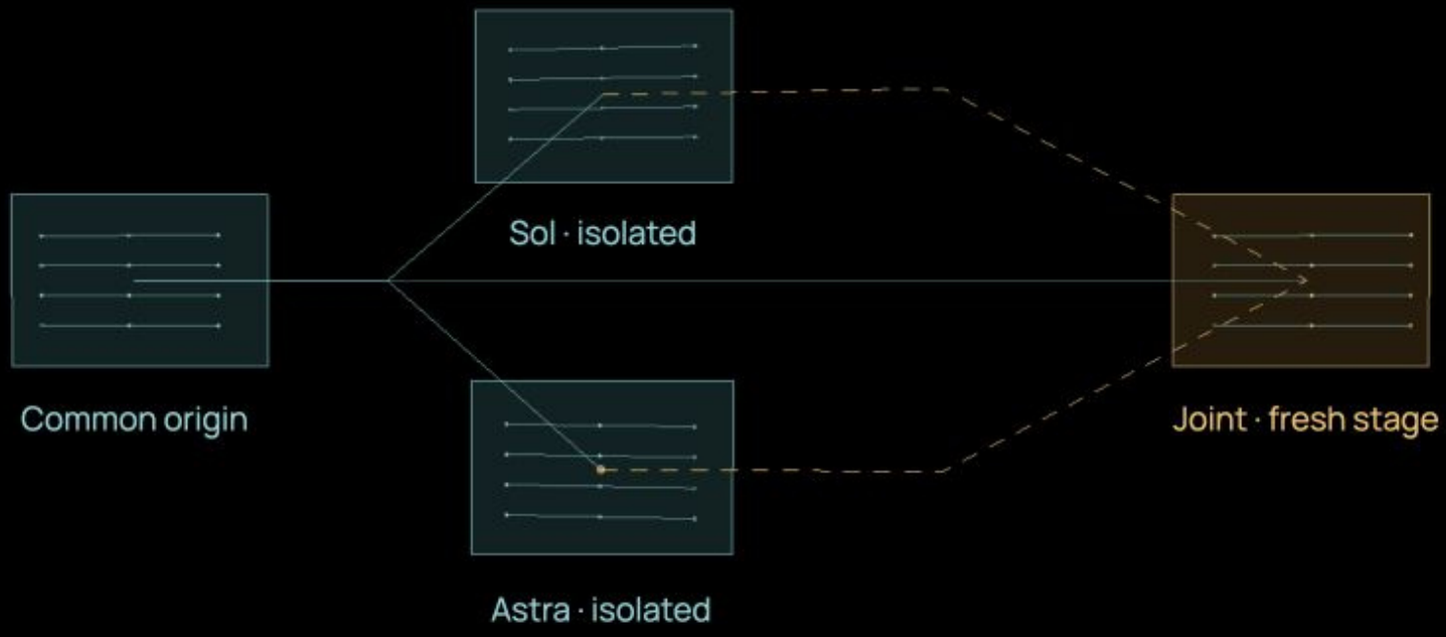
Real frozen weights · synthetic input illustration

After the edited sum passes through its rectifier and final output layer, the synthetic probability changes. This is a mechanistic illustration; saved held-out probability endpoints come later.

Selected Sol patch · 16 actual scalar changes in row 0 · column 15 has nonzero input activation in the fixed synthetic illustration.

Cooperation is a fresh branch

Five fresh joint rounds



Solid cyan: weights from origin · dashed amber: development history · joint starts fresh

The joint stage has five fresh rounds. It proposes complete interventions from the origin, rather than letting Astra modify a model already edited by Sol.

Two meanings of accuracy

Opposite movements are possible

Equal-class weighting ≠ equal-example weighting

$n_+ = 3{,}846$ $n_- = 12{,}435$ $N = 16{,}281$

Recall within a class = correct decisions / examples in that class

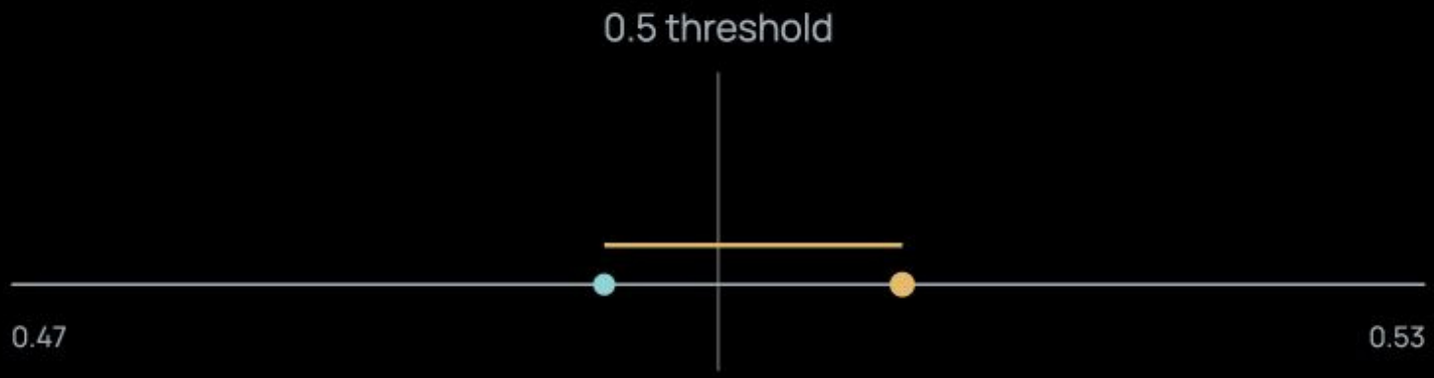
An edit can rescue positive examples while regressing more negative examples. Ordinary accuracy can decrease even when balanced accuracy increases, because their weighting rules differ.

Positive $n_+ = 3{,}846$ · negative $n_- = 12{,}435$ · $N = 16{,}281$. Recall = correct within that class / size of that class.

Two actual paired decisions

Row 77 · regressed by Sol

Anonymous row 77 · truth negative



Baseline p = 0.49516

Sol p = 0.50781

The selected Sol state moves this probability above one half too. That makes this decision wrong: a negative regression, illustrating the cost of the same directional tendency.

Saved anonymous Adult decisions · one-based rows 54 and 77 · p ≥ 0.5 predicts positive · no personal features or reconstructed activations.

Adult: the selected endpoints

A metric-specific improvement

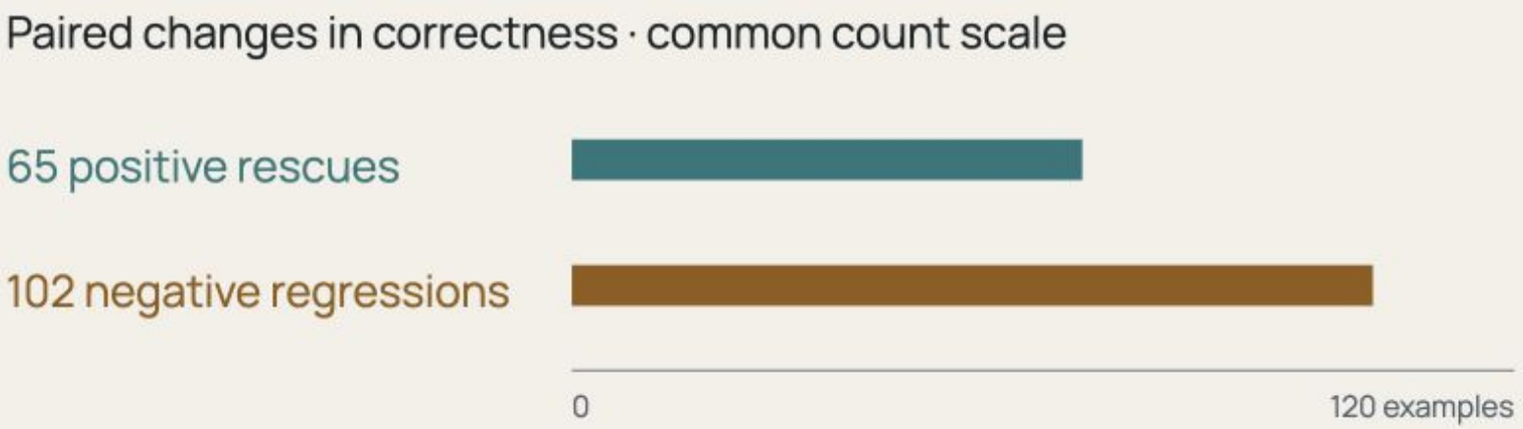
Selected state	Balanced accuracy	Ordinary accuracy
Damaged baseline	78.59%	82.86%
Sol	79.02%	82.63%
Astra	78.59%	82.80%
joint	78.97%	82.63%

The table supports a limited statement: Sol and joint improve balanced accuracy relative to this damaged baseline while losing ordinary accuracy. Uncertainty and controls determine how far that interpretation can travel.

Our held-out Adult results · 16,281 original test examples · BA = equal mean of class recalls; accuracy = total correct / N · main table rounded to 2 decimals; exact values in evidence.

Why those metrics move apart

ΔAccuracy = (65 − 102)/16,281



$$\Delta BA = \tfrac{1}{2}(65/3{,}846 - 102/12{,}435) = 0.4349 \text{ pp}$$

$$\Delta Accuracy = -37/16{,}281 = -0.2273 \text{ pp}$$

Ordinary accuracy instead counts a net loss of 37 correct decisions. The two metrics disagree for a transparent arithmetic reason, not because either has been relabeled.

Sol versus damaged baseline · raw paired example counts, not percentage points · class-weighted ΔBA uses original class denominators.

Controls keep the claim bounded

Damage criterion not met

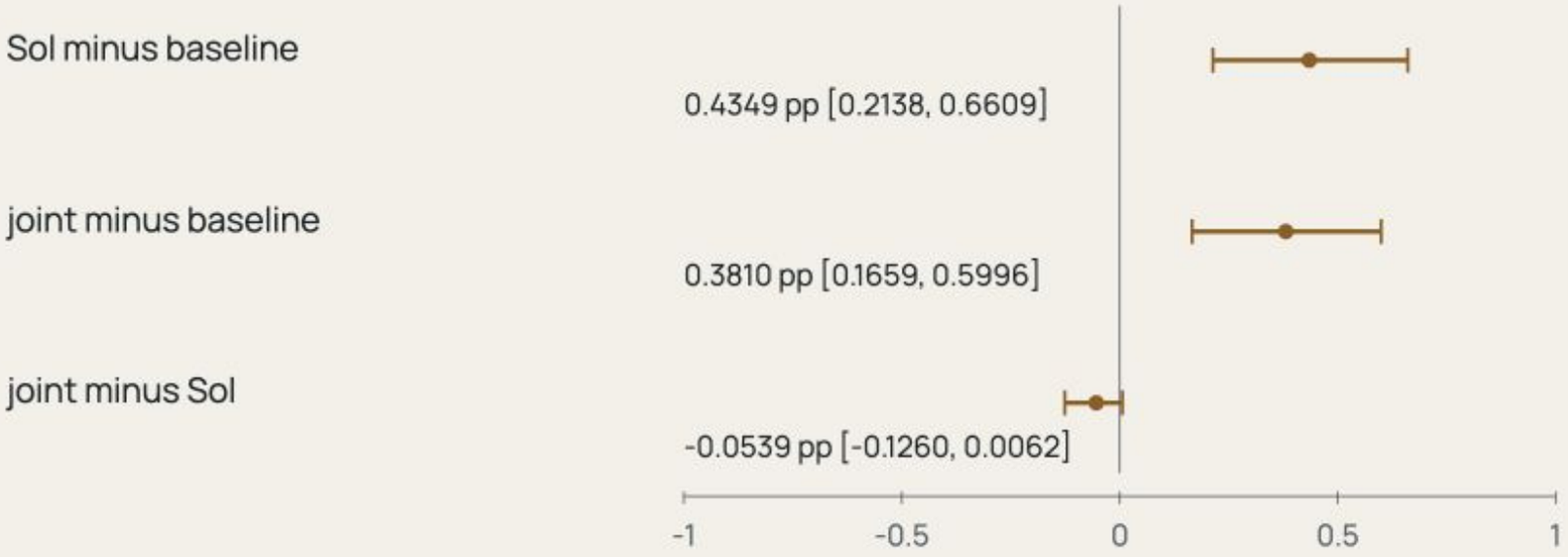
Frozen state	Balanced accuracy	Ordinary accuracy
Damaged baseline	78.59%	82.86%
clean	76.59%	85.63%
random	78.62%	82.81%
grid	78.57%	82.83%

The calibration damage criterion was insufficient for a conventional recovery interpretation. We therefore compare measured endpoints and avoid a recovery percentage with a misleading denominator.

Clean / random / grid controls retained · same frozen Adult test panel · insufficient_damage = true; no recovery percentage.

Uncertainty and cooperation

A panel is not a population of sessions



Difference in balanced accuracy · percentage points · 95% paired intervals

The intervals describe paired example uncertainty. They do not establish editor-session reliability, variability across seeds or a general benefit from cooperation.

Class-stratified paired bootstrap · seed101 · 2,000 draws · 95% intervals · percentage-point differences · example uncertainty, not session generalization.

Switch target: intact SmolLM2

Development and final panels

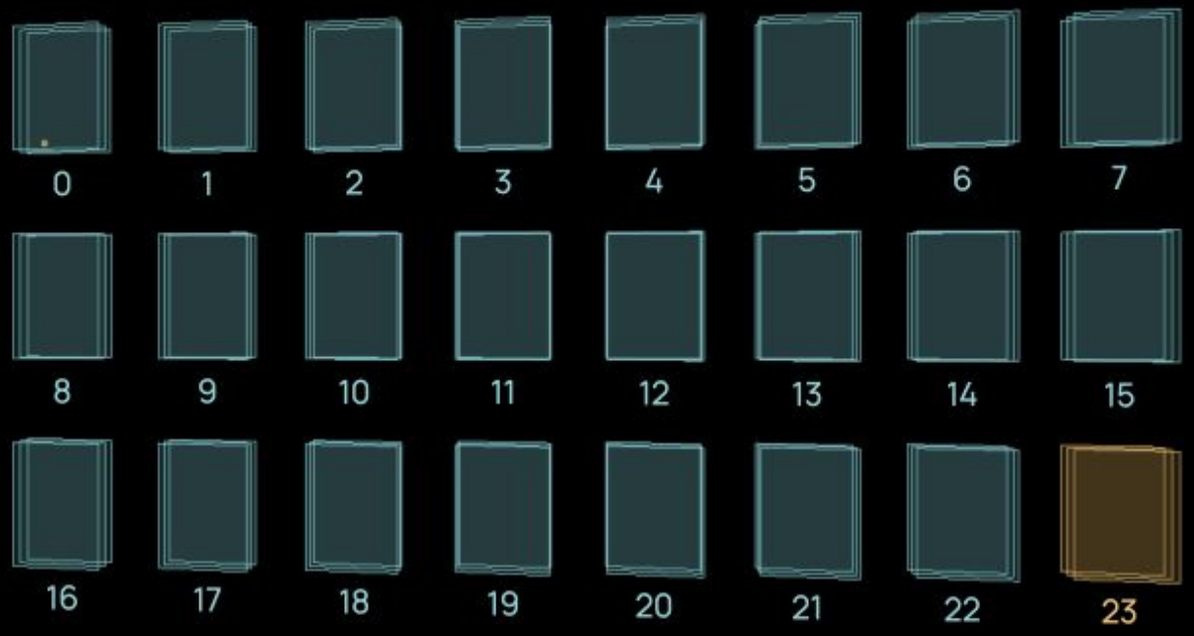


ARC final 512 · PIQA final 256

ARC uses training diagnostics and a 128-example development panel before the final 512 examples. The final PIQA panel has 256 examples; the model chooses completions, not census labels.

From a transformer to one block

24 transformer blocks



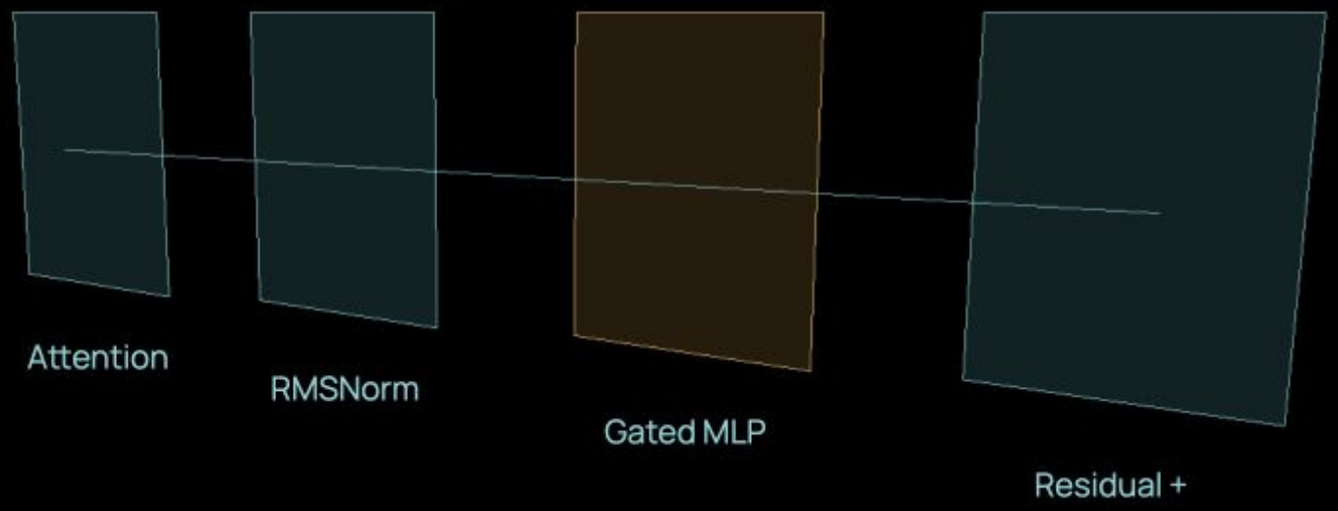
24 transformer blocks · zero-based indices · glass-sheet schematic

A transformer processes token representations through repeated blocks. The overview separates all 24 blocks so we can locate an intervention without turning the model into an unreadable stack.

Exact model architecture: 24 blocks · representative geometry, not a complete drawing of 1.7 billion coefficients.

From a transformer to one block

Focus on one block



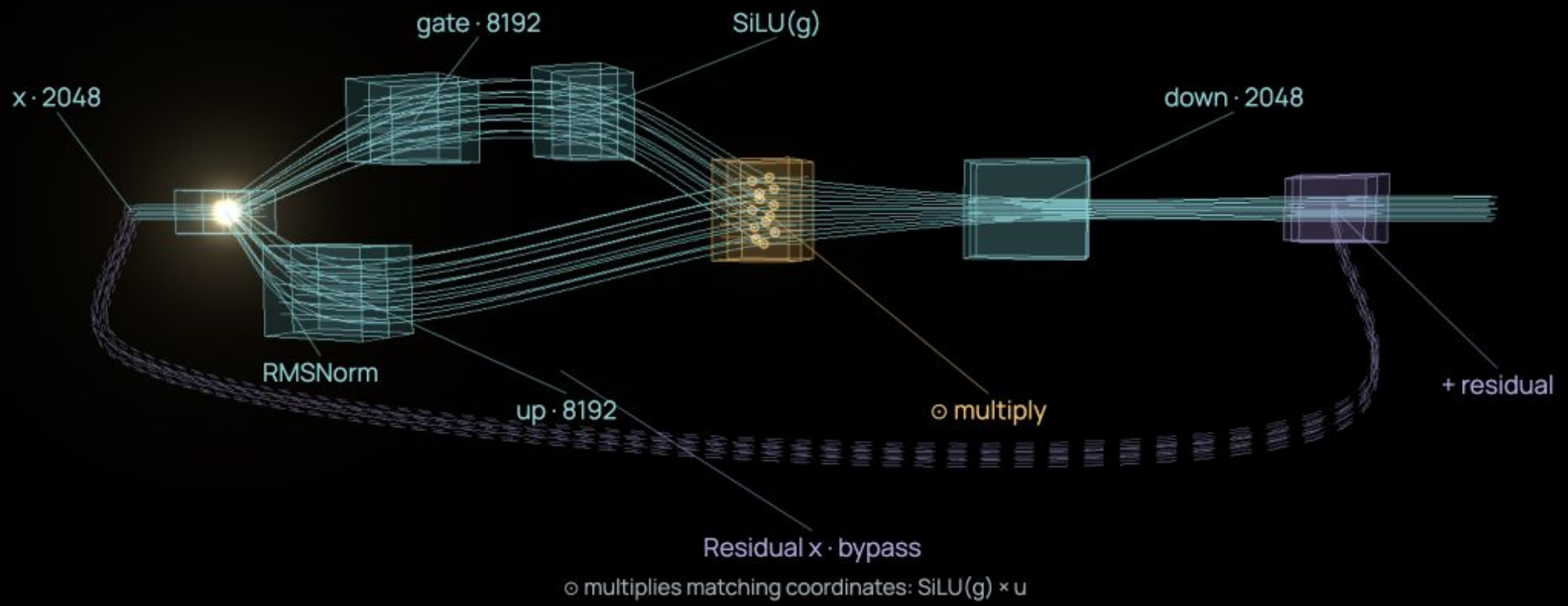
Inside final block 23 · attention and MLP both join the residual stream

Within a block, attention mixes information across tokens and a multilayer perceptron transforms each token's features. The intervention targets this feed-forward transformation.

Exact model architecture: 24 blocks · representative geometry, not a complete drawing of 1.7 billion coefficients.

From a transformer to one block

The multilayer perceptron

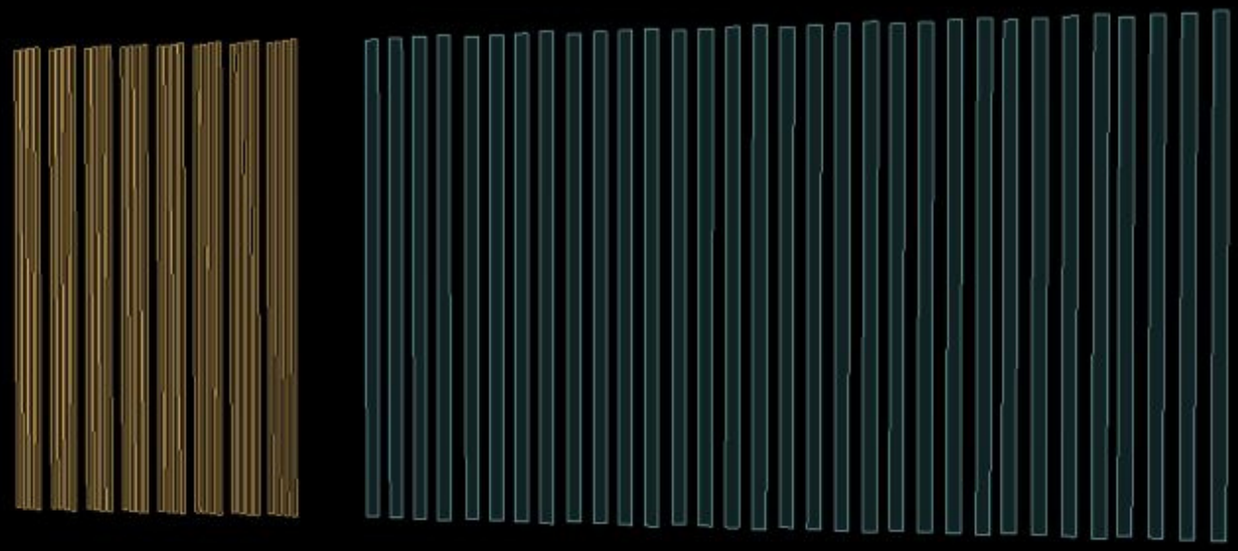


A multilayer perceptron, abbreviated MLP, expands a representation, gates its features and projects them back to the residual width. We will unpack each operation next.

Exact model architecture: 24 blocks · representative geometry, not a complete drawing of 1.7 billion coefficients.

From a transformer to one block

The down projection



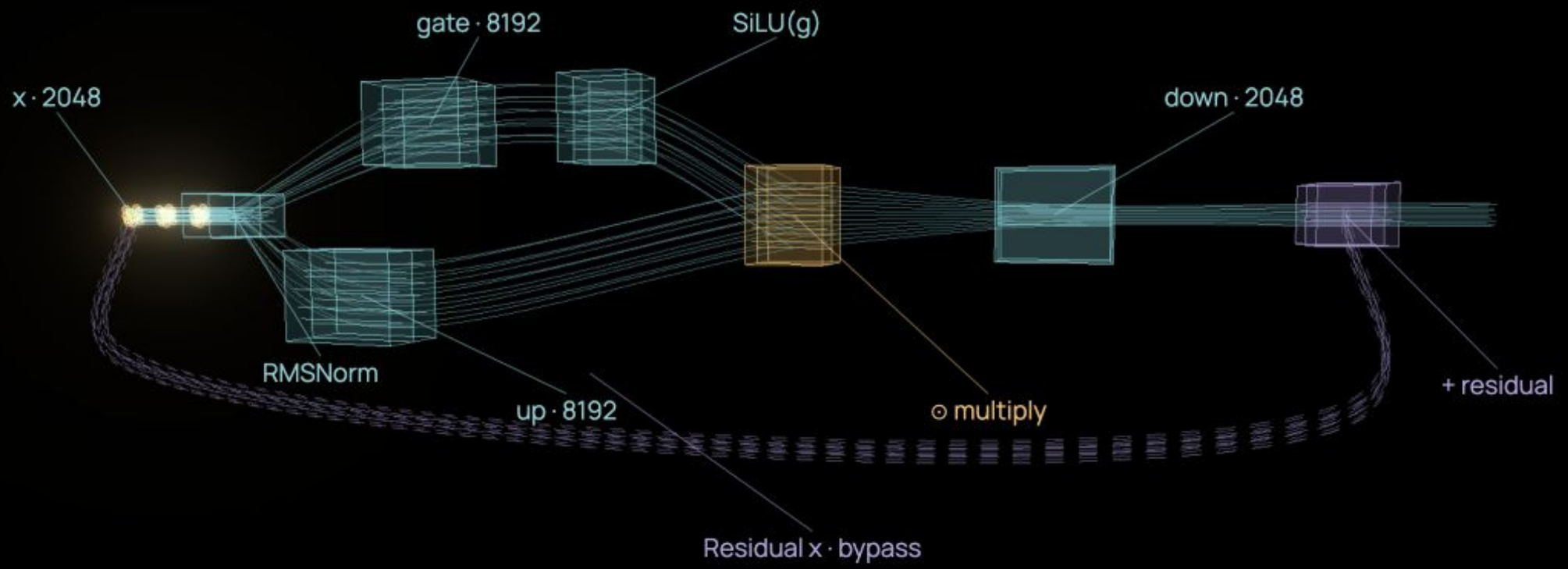
Representative down projection · 2048 rows × 8192 columns · scale columns or sparse delta

Only the down-projection weight tensor is eligible. Its columns connect expanded MLP features back into the model's representation, so column edits have a precise geometric meaning.

Exact model architecture: 24 blocks · representative geometry, not a complete drawing of 1.7 billion coefficients.

Inside the gated MLP

Root mean square normalization



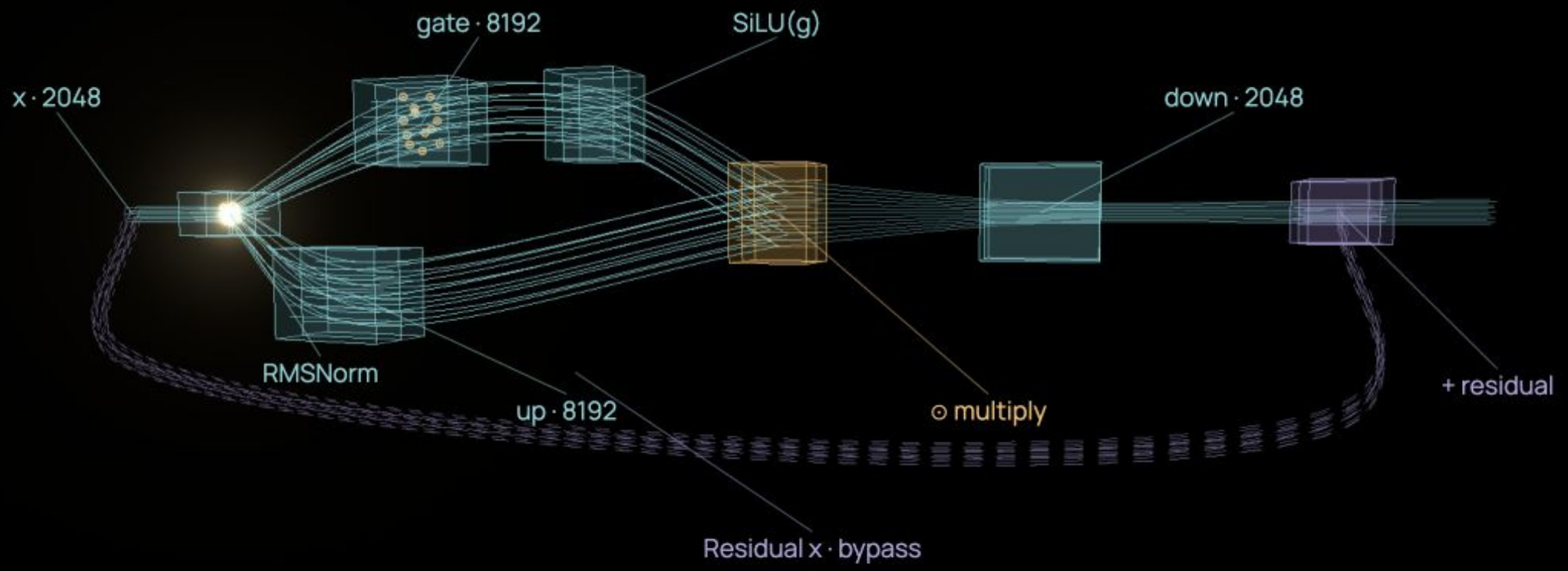
Tensor schematic · gate/up: 2048 → 8192 · down: 8192 → 2048

Root mean square normalization, or RMSNorm, rescales the incoming representation. The residual stream has width 2048; we denote the incoming residual by x and its normalized representation by $\hat{x}$.

Exact widths 2048 → 8192 → 2048 · schematic tensor geometry.

Inside the gated MLP

Gate and up projections



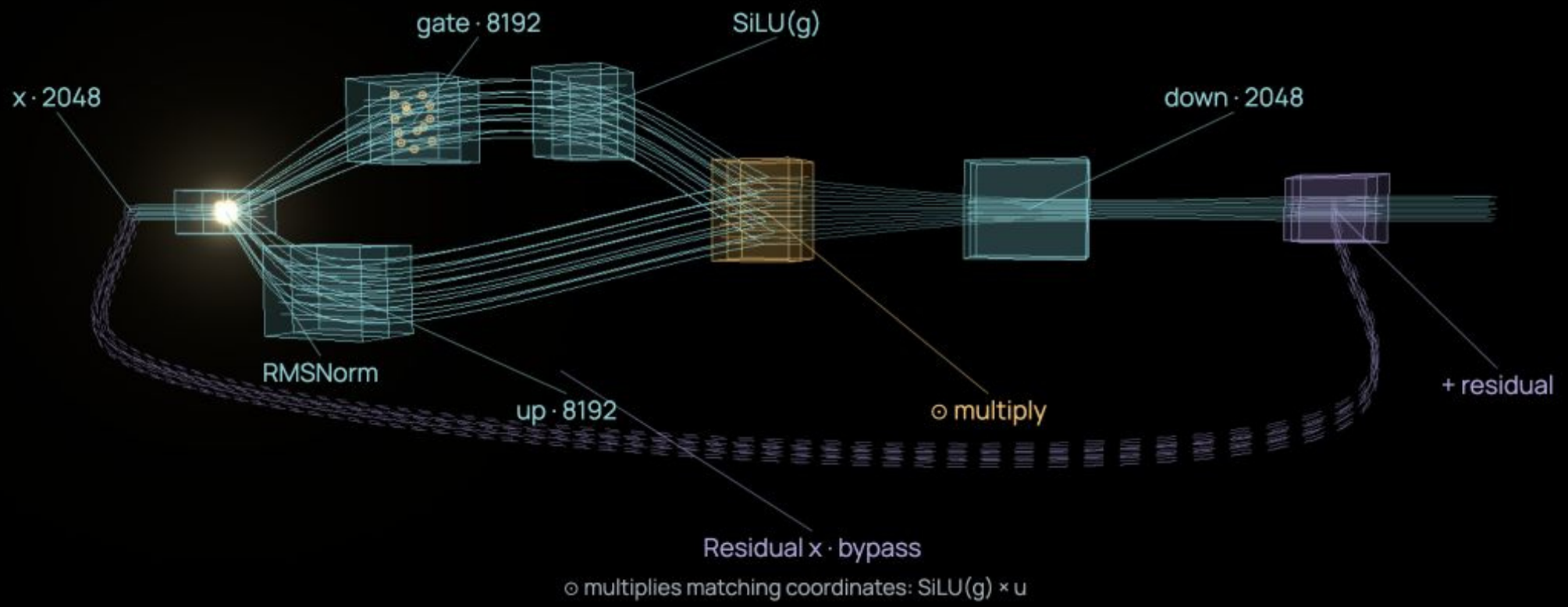
Tensor schematic · gate/up: 2048 → 8192 · down: 8192 → 2048

Two learned linear maps transform normalized $\hat{x}$ into 8192-dimensional vectors. Let g be the gate projection and u the up projection; they are different transforms of the same input.

Exact widths 2048 → 8192 → 2048 · schematic tensor geometry.

Inside the gated MLP

SiLU(g) ⊙ u

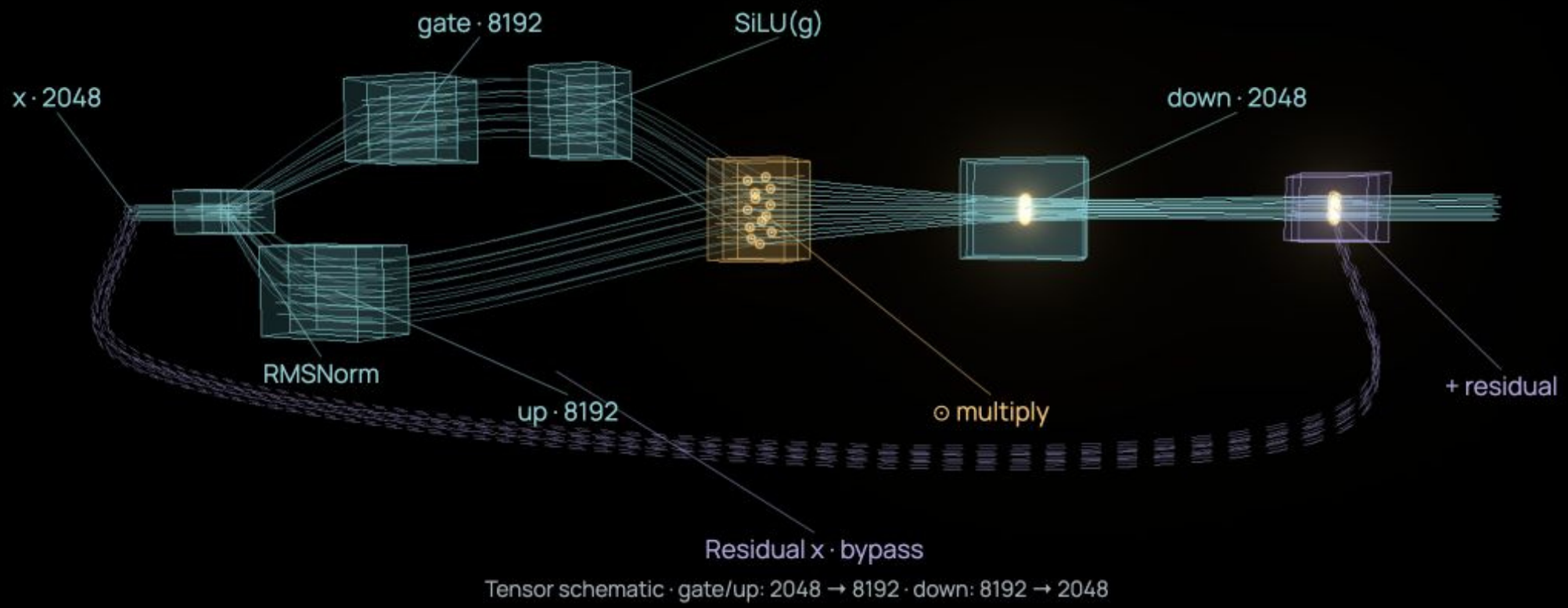


The sigmoid linear unit, or SiLU, maps g to g times its sigmoid. The symbol ⊙ means coordinate-by-coordinate multiplication with u, selecting the expanded feature contributions.

Exact widths 2048 → 8192 → 2048 · schematic tensor geometry.

Inside the gated MLP

Down projection to 2048

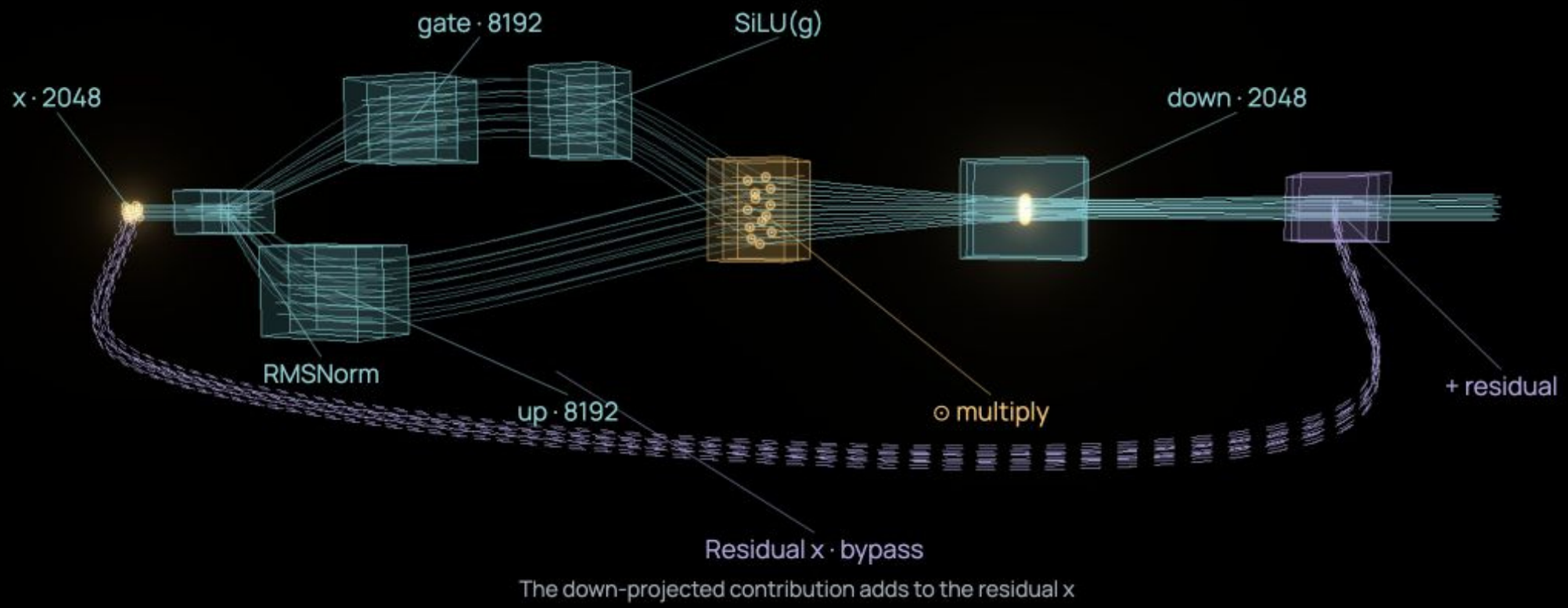


Tensor schematic · gate/up: 2048 → 8192 · down: 8192 → 2048

The down-projection matrix W_down maps the gated 8192 features back to 2048 dimensions. Selected columns are the only permitted intervention surface in this target.

Inside the gated MLP

Add back to the residual stream

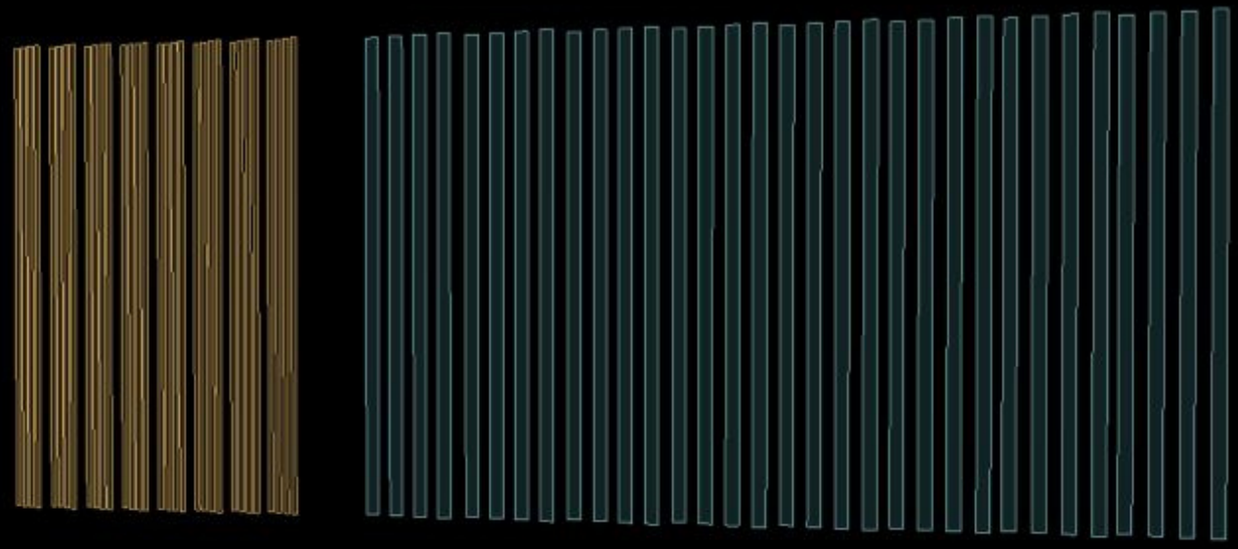


The projected change is added to the incoming residual representation. The bypass carries the original x around the MLP, so addition preserves a direct path for the incoming representation.

Exact widths 2048 → 8192 → 2048 · schematic tensor geometry.

Legal interventions in the language model

Geometry is representative

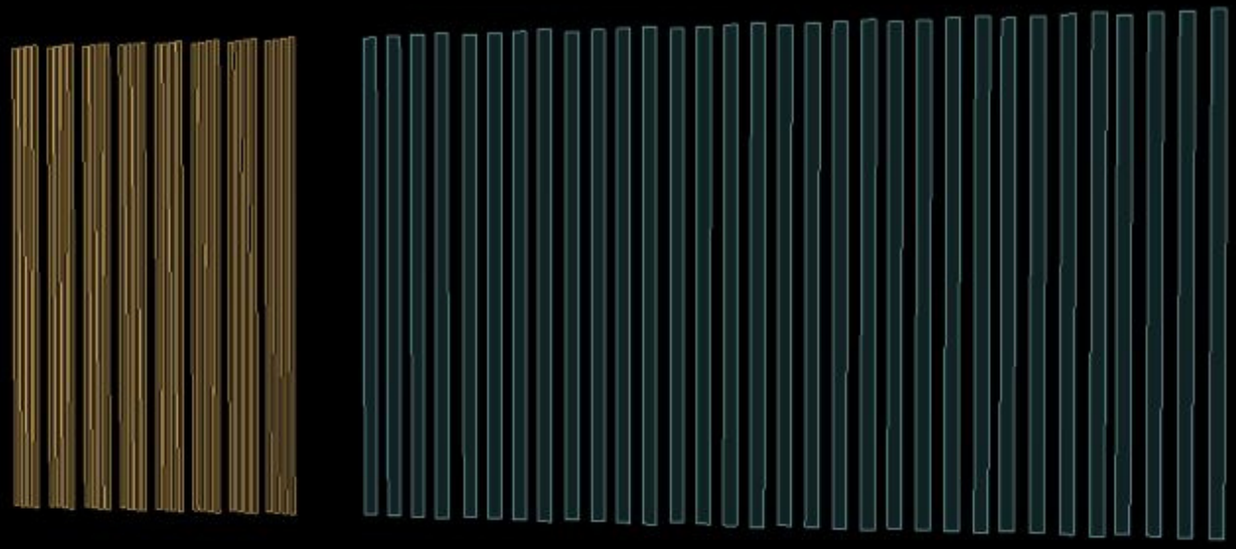


Representative down projection · 2048 rows × 8192 columns · scale columns or sparse delta

The highlighted columns identify the intervention pattern across all output rows. This tensor schematic locates the changes; candidate statistics describe their measured size.

A real proposal that was not selected

A proposal is not an endpoint



Astra candidate 1 · block 23 · columns 0…31 × 0.5 · 65,536 changes · L2 13.9107

The candidate is real and nonzero, but it did not win selection. A convincing visual intervention must not be mistaken for an improvement in the final selected model.

Actual Astra candidate 1 · final block index 23 · columns 0…31 × 0.5 · nonselected; no scalar samples available.

How an answer receives a score

A choice is a continuation

Illustrative continuation: “a rigid support”

Illustrative text continuation

Likelihoods are summed over continuation tokens; normalization uses choice characters

For each answer choice, the scorer asks how likely its continuation is given the question context. The example here is synthetic explanatory text, not a copied benchmark question.

How an answer receives a score

$L = \Sigma_t \log P(a_t \mid \text{context}, a_{<t})$

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t identifies a token; $a_{<t}$ means earlier tokens

Likelihoods are summed over continuation tokens; normalization uses choice characters

L denotes the sum of continuation token log probabilities. Token a_t is the current continuation token; $a_{<t}$ denotes the earlier continuation tokens already conditioned on.

How an answer receives a score

Raw choice = argmax L

Raw choice: $\text{argmax}_i L_i$

i identifies a choice; greatest sum wins

Likelihoods are summed over continuation tokens; normalization uses choice characters

Argmax means choose the index with the greatest score. Raw accuracy records whether this highest-likelihood choice matches the saved gold label; ties choose the first maximum.

How an answer receives a score

Normalized choice = argmax L / characters

Normalized choice: $\text{argmax}_i \, (L_i / c_i)$

c_i = number of characters in choice i

Likelihoods are summed over continuation tokens; normalization uses choice characters

Character-normalized accuracy divides by the answer text's character count before choosing. It does not divide by tokens, the context length or the number of choices.

The actual development trajectory

Equal score → smaller change



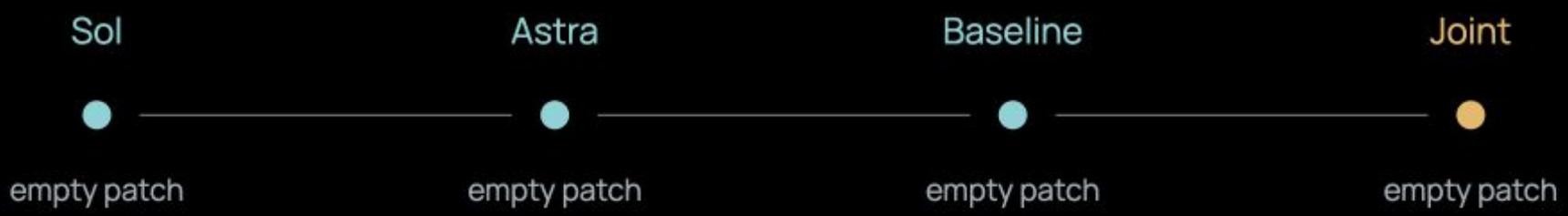
Candidate / attempt index · individual measured scores only · × = no score

Candidate zero has zero norm and zero changed parameters. When a nonzero proposal only ties its development score, the unchanged candidate wins the predefined tie-break.

Main corrected-prompt v2 · ARC development character-normalized accuracy · exact scored indices; no interpolated measurements.

Candidate zero wins

No selected improvement



One exact-weight group → shared predictions

Under these bounds and this development criterion, no edited candidate wins. That is the selection result; it neither erases the search nor proves that all possible edits are ineffective.

All four main v2 conditions select the unchanged origin · one exact-weight group · shared final predictions, not four independent observations.

Language-model final results

Baseline = Sol = Astra = joint

Task / convention	Correct / total	Accuracy
ARC-Easy · normalized	376 / 512	73.44%
ARC-Easy · raw	396 / 512	77.34%
PIQA · normalized	201 / 256	78.52%
PIQA · raw	205 / 256	80.08%

All four labels refer to the same selected weights and shared predictions. These main v2 measurements must not be confused with the older history in which PIQA failed before scoring.

Our completed main corrected-prompt v2 · ARC-Easy n = 512; PIQA n = 256 · character-normalized / raw continuation choice accuracy · identical across all conditions.

What the experiments say together

Bounded evidence, useful questions



One session · bounded development search

The outcomes differ across targets and selection surfaces. They motivate further controlled study without calling the null selection a failure or generalizing from one session to all weight-space editing.

One exploratory seed101 comparison · bounded interventions and development selection · no general editor-capability claim.

How to reproduce and interpret

Retain the design limits



Seed101 · Adult allowances 10 / 5

The original Adult solo allowance is ten and joint allowance five. Reproduction should preserve these methods, the character-based scorer and the frozen selection rule instead of silently changing the study.

Measurements answer the question

Thank you · discussion

Thank you · discussion

Davut Emre Taşar · BBVA AI Innovation

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